

Internet of Things and Vehicular Networks (TP-546)

IoT for Health: the Internet of Medical Things (IoMT)

Architectures, real cases, AI, 5G/6G, security and regulation — 2026 edition

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Outline

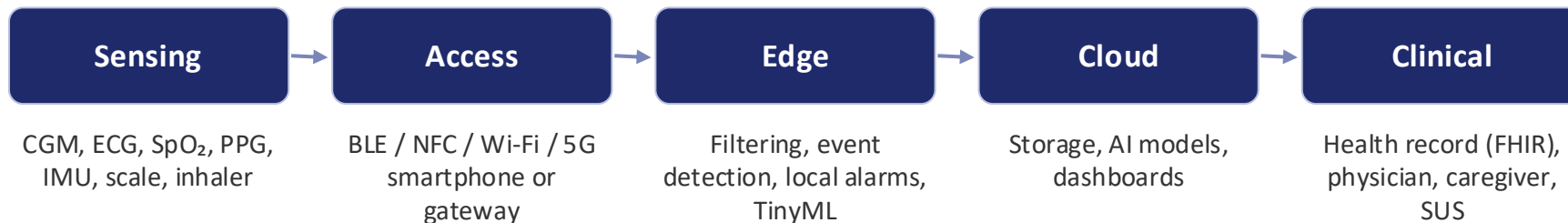
- 1 IoMT: concept, layered architecture and market
- 2 Communication and interoperability (BLE, 5G, IEEE 11073, HL7 FHIR)
- 3 Cases revisited: what worked, what failed and why
- 4 AI in IoMT: cleared devices, LLMs and edge AI
- 5 5G/6G and telesurgery: latency as a clinical requirement
- 6 Security and regulation (FDA, ANVISA, LGPD, EU)
- 7 Open research problems

What is IoMT?

Medical devices + connectivity + data analytics + clinical integration

- An ecosystem of sensors, medical devices, software and services that collect, transmit and analyze health data
- Classic benefits (still valid):
 - Continuous data instead of isolated measurements at the clinic
 - Remote care, fewer in-person visits, better treatment adherence
 - Patients with more control over their own condition
 - Clinical-grade devices sold over the counter (OTC CGM) and medical functions in consumer watches
 - Generative AI reading wearable data and medical records
 - Cybersecurity became a legal requirement for market access

Layered reference architecture



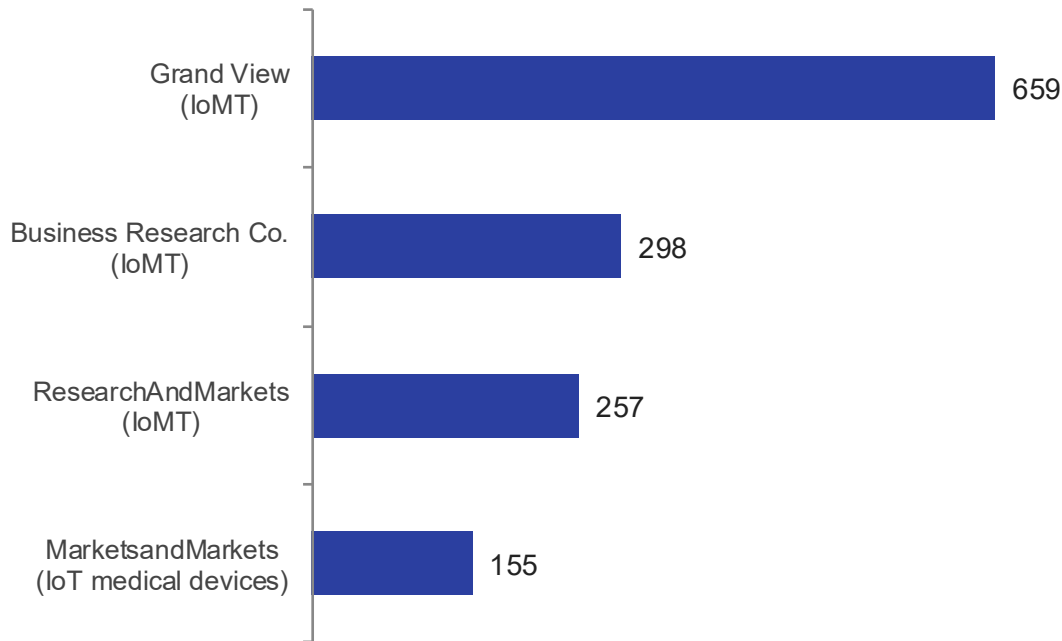
Cross-cutting concerns



Design question: at which layer should each clinical decision be taken, given the latency requirement and the risk?

Market: 2030 projections (US\$ billion)

Same topic, very different numbers — what does each report call “IoMT”?



- Roughly a 4× spread between estimates for the same year
- Reasons: scope (hardware only? software and services?), base year, proprietary methodology
- Consensus: growth of ~17% to 25% per year, driven by remote monitoring and wearables

Source: MarketsandMarkets (2025); ResearchAndMarkets/GlobeNewswire (Dec 2025); The Business Research Company (2026); Grand View Research (2026).

Communication technologies in IoMT

Technology	Range	Power	Typical healthcare use
BLE	~10–50 m	Very low	CGM, watches, insulin pens, scales
NFC	< 10 cm	Passive (sensor)	Sensor scanning (e.g. early FreeStyle Libre)
Wi-Fi	~50 m	High	Hospital monitors, home gateways
LoRaWAN / NB-IoT	km	Very low	Asset tracking, monitoring for the elderly
5G (eMBB/URLLC)	km	High	Connected ambulance, telesurgery, 4K video
6G (research)	—	—	Integrated sensing (ISAC), real-time digital twins

There is no single “best” option: the choice depends on latency, energy, patient mobility and the risk carried by the data.

Interoperability: the silent bottleneck

- **IEEE 11073 (PHD):** data model for personal health devices (pulse oximeter, scale, glucose meter)
- **Bluetooth GATT:** standardized health profiles (glucose, blood pressure, CGM, heart rate)
- **HL7 FHIR:** REST/JSON API to exchange clinical resources (Observation, Patient, Device) with health records
- **2026 case:** ChatGPT Health connects Epic and Oracle Health records through an intermediary (b.well) plus Apple Health data — it only works because standardized APIs exist
- **Research challenge:** semantic interoperability (same measurement, different units and context) and wearable data with no quality metadata

Source: HL7 FHIR R5; IEEE 11073-20601; OpenAI (Jan 2026, Jul 2026); Fierce Healthcare (Jul 2026).

Use Cases



Treatment follow-up: CYCORE

Randomized trial with head and neck cancer patients (2018)

- 357 patients under radiotherapy used a Bluetooth scale, a blood pressure monitor and a symptom-tracking app
- Data sent daily to the clinical team → early intervention on dehydration and side effects
- The monitored group had less severe symptoms than the group with conventional weekly visits
- Why it is a good example for graduate students:
 - It has a controlled clinical trial — the evidence is the differentiator, not the sensor
 - Simple off-the-shelf hardware: the value lies in the data flow up to the clinical decision

Digital pills: the rise and fall of Proteus

- **2017** FDA approves Abilify MyCite (Otsuka + Proteus): first drug with an ingestible sensor
- **2019** A US\$ 100M funding round falls through; most staff furloughed
- **Jun 2020** Proteus files for Chapter 11 bankruptcy
- **Aug 2020** Otsuka buys the assets for ~US\$ 15M (peak valuation was US\$ 1.5B)



Lessons

- The technology worked; the reimbursement model and the payer did not exist
- Sensor + patch + app + specialty pharmacy: too long a chain
- Ethical concerns about monitoring the patient (antipsychotic drug)
- Competitors carried on, e.g. etectRx ID-Cap (FDA 510(k), 2019)

Source: Fierce Healthcare (2020–2021); MobiHealthNews (2020); Fierce Biotech (Jun 2020).

Continuous glucose monitoring (CGM): 2024–2026

Over the counter (OTC)

Dexcom Stelo: first OTC CGM cleared by the FDA (Mar 2024), for adults not on insulin. Abbott Lingo and Libre Rio (Jun 2024). No hypoglycemia alarms.

One-year implant

Eversense 365: sensor implanted under the skin, a single warm-up per year; removable transmitter.

AID integration

Libre 3 Plus and Dexcom G7 feed automated insulin delivery pumps (t:slim X2, iLet, twiist).

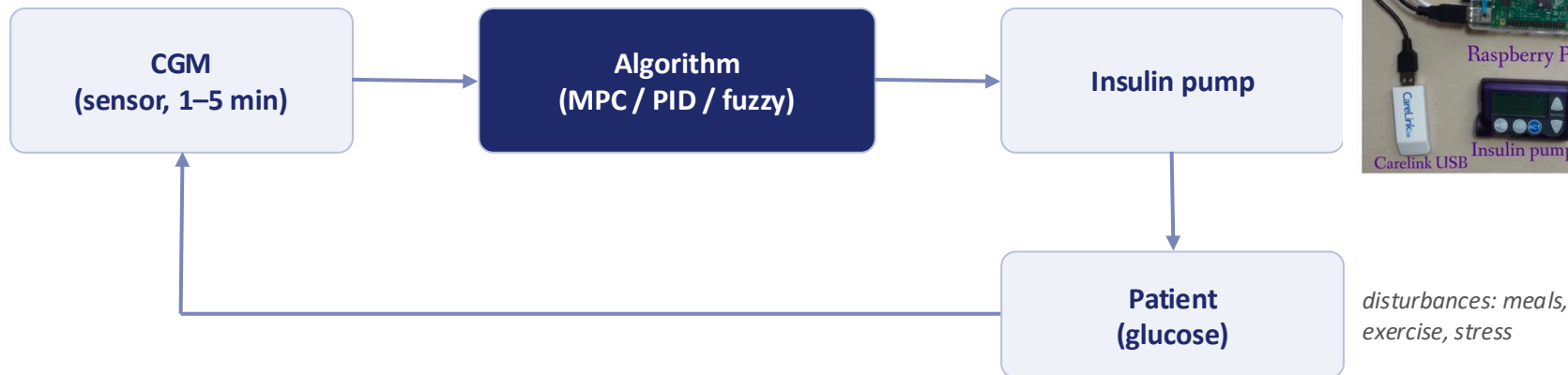
Quality at scale

Device correction on Libre 3 / 3 Plus batches in late 2025: manufacturing is a risk too.



Source: Dexcom/BusinessWire (Mar 2024); Drug Delivery Business (Aug 2024); GoodRx (Dec 2024); Type1Strong (2026).

Artificial pancreas: a closed-loop control problem



- **2015 — OpenAPS**: Dana Lewis and Scott Leibrand connect CGM + Raspberry Pi + pump (open source)
- **May 2023 — iLet (Beta Bionics)**: the user only states whether the meal is small, medium or large
- **Feb 2025 — Control-IQ+ (Tandem)**: FDA extends clearance to type 2 diabetes in adults, based on a randomized trial with >300 patients

Source: Tandem Diabetes Care (Feb 2025); Diabetesnet (2026).

Watches and earbuds with cleared medical functions

The line between wellness and medical device has become thinner

Year	Function (FDA clearance)	Sensor
2018	ECG / atrial fibrillation detection	electrical sensor
Sep 2024	Sleep apnea notifications (validated with 1,448 participants)	accelerometer
Sep 2024	AirPods Pro 2 as an OTC hearing aid	microphones + ML
Sep 2025	Hypertension notifications (30-day window; trained on >100,000 participants)	PPG + ML

- These are notifications, not diagnoses: the user is told to see a physician
- Research angle: can cheap sensors + ML + a huge user base replace dedicated medical sensors?

Source: MobiHealthNews (Sep 2025); Pulmonology Advisor (Sep 2024); Engadget (Sep 2025); Apple Newsroom (Sep 2024).

Inhalers, lenses and other cases: where we stand

Worked

- Connected inhalers (Propeller, FindAir): a sensor attached to the inhaler improves adherence
- Bluetooth coagulation testing (Roche, 2016): fewer clinic visits
- ResearchKit (Apple, 2018): movement disorder API for Parkinson's disease

Failed or pivoted

- Verily/Alcon glucose-sensing lens: shelved in 2018 — tear glucose was not reliable
- Proteus: technology cleared, company went bankrupt
- Lesson: measuring the right biomarker comes before the electronics

AI in IoMT

Cleared devices, LLMs reading health data and inference at the edge

AI-enabled devices authorized by the FDA

1,451

AI/ML devices on the FDA list by the end of 2025

295

authorized in 2025 alone (a record; 253 in 2024)

~76%

are radiology devices (Q4 2025: 55 of 72)

- As of Mar 2026, no device powered by generative AI or an LLM had been authorized
- JAMA study (691 devices): 46.7% of the summaries do not describe the study design and 53.3% omit the sample size
- Opportunity: AI on wearable signals (ECG, PPG, IMU) is still a minority on the list

Source: IntuitionLabs (2025–2026), based on the FDA public list; The Medical Futurist (Mar 2026).

LLMs + wearable and medical record data

- **Jan 2026:** OpenAI announces ChatGPT Health (beta): a separate space connecting Apple Health, wellness apps and medical records
- **Jul 2026:** released to adults in the US; records via Epic and Oracle Health
- **Scale:** more than 300 million people per week ask ChatGPT health questions (company figure)
- **Positioning:** it does not diagnose or treat; health data does not train the base models by default
- Questions for discussion:
 - Who validates the interpretation of a noisy wearable measurement?
 - Consent, stale records and user over-reliance
 - How would this be classified under ANVISA RDC 657/2022 (SaMD) and the LGPD in Brazil?

Source: OpenAI (Jan and Jul 2026); CNBC (Jan 2026); Fierce Healthcare (Jul 2026); MacRumors (Jul 2026).

Edge AI: where should the model run?

Location	Latency	Power	Privacy	Example
Sensor (MCU, TinyML)	ms	μW – mW	Highest	On-device arrhythmia or fall detection
Smartphone / gateway	10–100 ms	mW – W	High	Sleep staging, CGM alerts
Edge (Jetson, 5G MEC)	10–50 ms	W	Medium	ICU video, ambulance telemetry
Cloud	100 ms–s	—	Depends	LLMs, population models, digital twins

- Federated learning: train across hospitals without moving the data
- Trade-off worth researching: accuracy $\times$ power $\times$ memory $\times$ explainability

Order-of-magnitude values, for comparison only.

Digital twins in healthcare

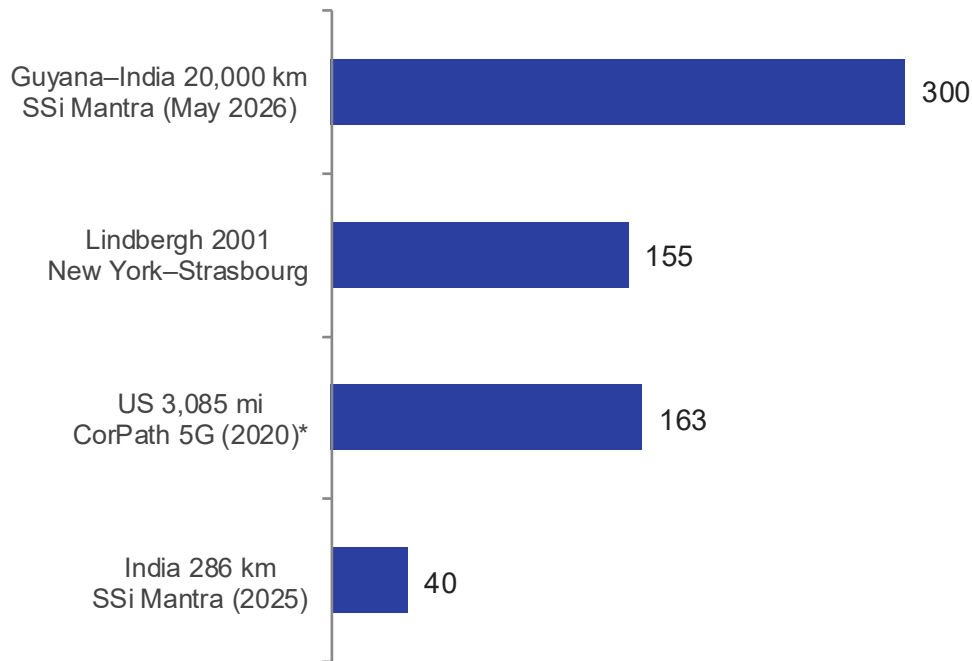
- A virtual model of a patient or organ, synchronized with real IoMT data to predict and simulate interventions
- Examples:
 - Siemens Healthineers and Dassault (Living Heart): cardiac models
 - ICU: a glycemia digital twin for septic patients built on a time-series foundation model, personalized in seconds on a standard laptop (medRxiv, 2026)
 - NSF / Mayo Clinic project: causal AI digital twin for critical care
- Open challenges:
 - Validation and prediction uncertainty; continuous updating with missing data
 - Network requirements for real-time synchronization — a direct link to 6G

Source: medRxiv 10.64898/2026.04.24.26351177; NSF IIS-2123848; Laamarti et al. (arXiv 2301.03930).

5G/6G and telesurgery

When network latency becomes a clinical requirement

Robotic telesurgery: distance × latency



- May 2026: cardiac surgery (internal mammary artery takedown) with the surgeon in Guyana and the patient in India, over fiber, at 290–300 ms
- Over regional distances, 35–40 ms has already been achieved
- *CorPath: vascular intervention on a simulator, not on a patient
- Jitter and packet loss cause mechanical tremor — as important as the mean latency

Source: SS Innovations/SEC (2026); MassDevice (Jun 2026); CBEHx (2026); Marescaux et al., Ann Surg (2002); protocol NCT07132255.

Security and regulation

An insecure device is a clinically unsafe device

Real case: backdoor in a patient monitor (Jan 2025)

- CISA and the FDA warn that the Contec CMS8000 monitor (also relabeled as Epsimed MN-120) contains a backdoor in its firmware
- **CVE-2025-0626**: connection to a hard-coded IP address, allowing files to be downloaded and executed without verification
- **CVE-2025-0683**: patient data sent in plain text to that IP address
- **CVE-2024-12248**: out-of-bounds write over UDP → remote code execution (CVSS v3.1 9.8)
- Official recommendation: remove the monitors from the network — no patch was available
- Design lessons: SBOM, third-party firmware inspection, network segmentation and outbound traffic monitoring

Source: CISA ICSMA-25-030-01 and fact sheet (Jan–Feb 2025); FDA Safety Communication; NHS England Digital (2025).

Open research problems

Possible topics for a dissertation or thesis

Energy

Energy harvesting and low-power protocols for implantable and long-wear sensors

Lightweight security

Cryptography (including post-quantum) and secure boot on resource-constrained MCUs

Trustworthy AI

Clinical validation, bias, uncertainty and explainability on wearable signals

Interoperability

Mapping sensor data to FHIR with semantics and data quality

Networking

Latency/jitter guarantees for telesurgery and safe degradation

Privacy

Federated learning and differential privacy across hospitals

Videos

- How doctors can help AI to revolutionize medicine — <https://youtu.be/rCkXW079cDM>
- How IoT Will Impact Healthcare — Brian Weldy (HCA) — <https://youtu.be/jbbxZ52PCEc>
- David Moss: Smart aging — IoT, the elderly and caregivers — <https://youtu.be/H00TqDbBKL4>
- FindAir is a smart inhaler — <https://youtu.be/yWlQ63l41aw>
- Triggerfish: Contact Lens-based Glaucoma Solution — <https://youtu.be/9cm7ljKPhmo>
- Eversense Changes the Lives of Those with Diabetes — https://youtu.be/3BQKU0e_MhA
- Revolutionizing glucose monitoring with FreeStyle Libre — <https://youtu.be/mdK5640AfwA>

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